

GE Healthcare

Three Phase Partial UPS Installation

For use with:

PET Discovery and CT Lightspeed Family

OPERATING DOCUMENTATION



5272751-1EN

Rev 1

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IMPORTANT PRECAUTIONS

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.
- V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKLADU DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ Vlivem ELEKTRICKÉHO PROUDU, RESPEKTIVE Vlivem MECHANICKÝCH ČI JINÝCH RIZIK.

ADVARSEL

(DA)

- DENNE SERVICE MANUAL FINDES KUN PÅ ENGELSK.
- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FØRSØG IKKE AT SERVICE UDSTYRET MEDMINDRE DENNE SERVICE MANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING

(NL)

- DEZE ONDERHOUDSHANDLEIDING IS ENKEL IN HET ENGELS VERKRIJGBAAR.
- ALS HET ONDERHOUDSPERSONEEL EEN ANDERE TAAL VEREIST, DAN IS DE KLANT VERANTWOORDELIJK VOOR DE VERTALING ERVAN.
- PROBEER DE APPARATUUR NIET TE ONDERHOUDEN VOORDAT DEZE ONDERHOUDSHANDLEIDING WERD GERAADPLEEGD EN BEGREPEN IS.
- INDIEN DEZE WAARSCHUWING NIET WORDT OPGEVOLGD, ZOU HET ONDERHOUDSPERSONEEL, DE OPERATOR OF EEN PATIËNT GEWOND KUNNEN RAKEN ALS GEVOLG VAN EEN ELEKTRISCHE SCHOK, MECHANISCHE OF ANDERE GEVAREN.

WARNING

(EN)

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR, OR PATIENT FROM ELECTRIC SHOCK, OR FROM MECHANICAL OR OTHER HAZARDS.

HOIATUS

(ET)

- KÄESOLEV TEENINDUSJUHEND ON SAADAVAL AINULT INGLISE KEELES.
- KUI KLIENDITEENINDUSE OSUTAJA NÕUAB JUHENDIT INGLISE KEELEST ERINEVAS KEELES, VASTUTAB KLIENT TÕLKETEENUSE OSUTAMISE EEST.
- ÄRGE ÜRITAGE SEADMEID TEENINDADA ENNE EELNEVALT KÄESOLEVA TEENINDUSJUHENDIGA TUTVUMIST JA SELLEST ARU SAAMIST.

VAROITUS

(FI)

- KÄESOLEVA HOIATUSE EIRAMINE VÖIB PÖHJUSTADA TEENUSEOSUTAJA, OPERAATORI VÖI PATSIENDI VIGASTAMIST ELEKTRILÖÖGI, MEHAANILISE VÖI MUU OHU TAGAJÄRJEL.
- TÄMÄ HUOLTO-OHJE ON SAATAVILLA VAIN ENGLANNIKSI.
- JOS ASIAKKAAN HUOLTOHENKILÖSTÖ VAATII MUUTA KUIN ENGLANNINKIELISTÄ MATERIAALIA, TARVITTAVAN KÄÄNNÖKSEN HANKKIMINEN ON ASIAKKAAN VASTUULLA.
- ÄLÄ YRITÄ KORJATA LAITTEISTOA ENNEN KUIN OLET VARMASTI LUKENUT JA YMMÄRTÄNYT TÄMÄN HUOLTO-OHJEEN.
- MIKÄLI TÄTÄ VAROITUSTA EI NOUDATETA, SEURAUKSENA VOI OLLA HUOLTOHENKILÖSTÖN, LAITTEISTON KÄYTTÄJÄN TAI POTILAAN VAHINGOITTUMINEN SÄHKÖISKUN, MEKAANISEN VIAN TAI MUUN VAARATILANTEEN VUOKSI.

ATTENTION

(FR)

- CE MANUEL DE SERVICE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENIR SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

(DE)

- DIESE SERVICEANLEITUNG EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT DIESE ANLAGE ZU WARTEN, OHNE DIESE SERVICEANLEITUNG GELESEN UND VERSTANDEN ZU HABEN.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH STROMSCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ ΔΙΑΤΙΘΕΤΑΙ ΣΤΑ ΑΓΓΛΙΚΑ ΜΟΝΟ.
- ΕΑΝ ΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ ΕΝΟΣ ΠΕΛΑΤΗ ΑΠΑΙΤΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕ ΓΛΩΣΣΑ ΕΚΤΟΣ ΤΩΝ ΑΓΓΛΙΚΩΝ, ΑΠΟΤΕΛΕΙ ΕΥΘΥΝΗ ΤΟΥ ΠΕΛΑΤΗ ΝΑ ΠΑΡΕΧΕΙ ΥΠΗΡΕΣΙΕΣ ΜΕΤΑΦΡΑΣΗΣ.
- ΜΗΝ ΕΠΙΧΕΙΡΗΣΕΤΕ ΤΗΝ ΕΚΤΕΛΕΣΗ ΕΡΓΑΣΙΩΝ ΣΕΡΒΙΣ ΣΤΟΝ ΕΞΟΠΛΙΣΜΟ ΕΚΤΟΣ ΕΑΝ ΕΧΕΤΕ ΣΥΜΒΟΥΛΕΥΤΕΙ ΚΑΙ ΕΧΕΤΕ ΚΑΤΑΝΟΗΣΕΙ ΤΟ ΠΑΡΟΝ ΕΓΧΕΙΡΙΔΙΟ ΣΕΡΒΙΣ.
- ΕΑΝ ΔΕ ΛΑΒΕΤΕ ΥΠΟΨΗ ΤΗΝ ΠΡΟΕΙΔΟΠΟΙΗΣΗ ΑΥΤΗ, ΕΝΔΕΧΕΤΑΙ ΝΑ ΠΡΟΚΛΗΘΕΙ ΤΡΑΥΜΑΤΙΣΜΟΣ ΣΤΟ ΑΤΟΜΟ ΠΑΡΟΧΗΣ ΣΕΡΒΙΣ, ΣΤΟ ΧΕΙΡΙΣΤΗ Ή ΣΤΟΝ ΑΣΘΕΝΗ ΑΠΟ ΗΛΕΚΤΡΟΠΛΗΞΙΑ, ΜΗΧΑΝΙΚΟΥΣ Ή ΑΛΛΟΥΣ ΚΙΝΔΥΝΟΥΣ.

FIGYELMEZTETÉS

(HU)

- EZEN KARBANTARTÁSI KÉZIKÖNYV KIZÁRÓLAG ANGOL NYELVEN ÉRHETŐ EL.
- HA A VEVŐ SZOLGÁLTATÓJA ANGOLTÓL ELTÉRŐ NYELVRE TART IGÉNYT, AKKOR A VEVŐ FELELŐSSÉGE A FORDÍTÁS ELKÉSZÍTTETÉSE.
- NE PRÓBÁLJA ELKEZDENI HASZNÁLNI A BERENDEZÉST, AMÍG A KARBANTARTÁSI KÉZIKÖNYVBEN LEÍRTAKAT NEM ÉRTELMEZTÉK.
- EZEN FIGYELMEZTETÉS FIGYELMEN KÍVÜL HAGYÁSA A SZOLGÁLTATÓ, MŰKÖDTETŐ VAGY A BETEG ÁRAMÚTÉS, MECHANIKAI VAGY EGYÉB VESZÉLYHELYZET MIATTI SÉRÜLÉSÉT EREDMÉNYEZHETI.

AÐVÖRUN

(IS)

- ÞESSI ÞJÓNUSTUHANDBÓK ER EINGÖNGU FÁANLEG Á ENSKU.
- EF AÐ ÞJÓNUSTUVEITANDI VIÐSKIPTAMANNS ÞARFNAST ANNAS TUNGUMÁLS EN ENSKU, ER ÞAÐ SKYLDA VIÐSKIPTAMANNS AÐ SKAFFA TUNGUMÁLAÞJÓNUSTU.
- REYNIÐ EKKI AÐ AFGREIÐA TÆKIÐ NEMA AÐ ÞESSI ÞJÓNUSTUHANDBÓK HEFUR VERIÐ SKOÐUÐ OG SKILIN.
- BROT Á SINNA ÞESSARI AÐVÖRUN GETUR LEITT TIL MEIÐSLA Á ÞJÓNUSTUVEITANDA, STJÓRNANDA EÐA SJÚKLINGS FRÁ RAFLOSTI, VÉLRÆNU EÐA ÖÐRUM ÁHÆTTUM.

AVVERTENZA

(IT)

- IL PRESENTE MANUALE DI MANUTENZIONE E DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE E TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO
- IL NON RISPETTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다.
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.
- 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRĪDINĀJUMS

(LV)

- ŠĪ APKALPES ROKASGRĀMATA IR PIEEJAMA TIKAI ANGLŪ VALODĀ.
- JA KLIENTA APKALPES SNIEDZĒJAM NEPIECIEŠAMA INFORMĀCIJA CITĀ VALODĀ, NEVIS ANGLŪ, KLIENTA PIENĀKUMS IR NODROŠINĀT TULKOŠANU.
- NEVEICIET APRĪKOJUMA APKALPI BEZ APKALPES ROKASGRĀMATAS IZLASĪŠANAS UN SAPRAŠANAS.
- ŠĪ BRĪDINĀJUMA NEIEVĒROŠANA VAR RADĪT ELEKTRISKĀS STRĀVAS TRIECIENA, MEHĀNISKU VAI CITU RISKU IZRAISĪTU TRAUMU APKALPES SNIEDZĒJAM, OPERATORAM VAI PACIENTAM.

ĮSPĖJIMAS

(LT)

- ŠIS EKSPLOATAVIMO VADOVAS YRA PRIEINAMAS TIK ANGLŲ KALBA.
- JEI KLIENTO PASLAUGŲ TIEKĒJAS REIKALAUJA VADOVO KITA KALBA – NE ANGLŲ, NUMATYTI VERTIMO PASLAUGAS YRA KLIENTO ATSAKOMYBĖ.
- NEMĖGINKITE ATLIKTI ĮRANGOS TECHNINĖS PRIEŽIŪROS, NEBENT ATSIŽVELGĖTE Į ŠĮ EKSPLOATAVIMO VADOVĄ IR JĮ SUPRATOTE.
- JEI NEATKREIPSITE DĖMESIO Į ŠĮ PERSPĖJIMĄ, GALIMI SUŽALOJIMAI DĖL ELEKTROS ŠOKO,
- MECHANINIŲ AR KITŲ PAVOJŲ PASLAUGŲ TIEKĖJUI, OPERATORIUI AR PACIENTUI.

ADVARSEL

(NO)

- DENNE SERVICEHÅNDBOKEN FINNES BARE PÅ ENGELSK.
- HVIS KUNDENS SERVICELEVERANDØR TRENGER ET ANNET SPRÅK, ER DET KUNDENS ANSVAR Å SØRGE FOR OVERSETTELSE.
- IKKE FORSØK Å REPARERE UTSTYRET UTEN AT DENNE SERVICEHÅNDBOKEN ER LEST OG FORSTÅTT.
- MANGLENDE HENSYN TIL DENNE ADVARSELEN KAN FØRE TIL AT SERVICELEVERANDØREN, OPERATØREN ELLER PASIENTEN SKADES PÅ GRUNN AV ELEKTRISK STØT, MEKANISKE ELLER ANDRE FARER.

OSTRZEŻENIE

(PL)

- NINIEJSZY PODRĘCZNIK SERWISOWY DOSTĘPNY JEST JEDYNIEM W JĘZYKU ANGIELSKIM.
- JEŚLI DOSTAWCA USŁUG KLIENTA WYMAGA JĘZYKA INNEGO NIŻ ANGIELSKI, ZAPEWNIENIE USŁUGI TŁUMACZENIA JEST OBOWIĄZKIEM KLIENTA.
- NIE PRÓBOWAĆ SERWISOWAĆ WYPOSAŻENIA BEZ ZAPOZNANIA SIĘ I ZROZUMIENIA NINIEJSZEGO PODRĘCZNIKA SERWISOWEGO.
- NIEZASTOSOWANIE SIĘ DO TEGO OSTRZEŻENIA MOŻE SPOWODOWAĆ URAZY DOSTAWCY USŁUG, OPERATORA LUB PACJENTA W WYNIKU PORAŻENIA ELEKTRYCZNEGO, ZAGROŻENIA MECHANICZNEGO BĄDŹ INNEGO.

ATENÇÃO

(PT-BR)

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENÇÃO

(PT-PT)

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA
- O NÃO CUMPRIMENTO DESTE AVISO PODE COLOCAR EM PERIGO A SEGURANÇA DO TÉCNICO, DO OPERADOR OU DO PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

ATENȚIE

(RO)

- ACEST MANUAL DE SERVICE ESTE DISPONIBIL NUMAI ÎN LIMBA ENGLEZĂ.
- DACĂ UN FURNIZOR DE SERVICII PENTRU CLIEȚI NECESITĂ O ALTĂ LIMBĂ DECÂT CEA ENGLEZĂ, ESTE DE DATORIA CLIENTULUI SĂ FURNIZEZE O TRADUCERE.
- NU ÎNCERCAȚI SĂ REPARAȚI ECHIPAMENTUL DECÂT ULTERIOR CONSULTĂRII ȘI ÎNȚELEGERII ACESTUI MANUAL DE SERVICE.
- IGNORAREA ACESTUI AVERTISMENT AR PUTEA DUCE LA RĂNIREA DEPANATORULUI, OPERATORULUI SAU PACIENTULUI ÎN URMA PERICOLELOR DE ELECTROCUTARE, MECANICE SAU DE ALTĂ NATURĂ.

ОСТОРОЖНО!

(RU)

- ДАННОЕ РУКОВОДСТВО ПО ОБСЛУЖИВАНИЮ ПРЕДЛАГАЕТСЯ ТОЛЬКО НА АНГЛИЙСКОМ ЯЗЫКЕ.
- ЕСЛИ СЕРВИСНОМУ ПЕРСОНАЛУ КЛИЕНТА НЕОБХОДИМО РУКОВОДСТВО НЕ НА АНГЛИЙСКОМ, А НА КАКОМ-ТО ДРУГОМ ЯЗЫКЕ, КЛИЕНТУ СЛЕДУЕТ САМОСТОЯТЕЛЬНО ОБЕСПЕЧИТЬ ПЕРЕВОД.
- ПЕРЕД ОБСЛУЖИВАНИЕМ ОБОРУДОВАНИЯ ОБЯЗАТЕЛЬНО ОБРАТИТЕСЬ К ДАННОМУ РУКОВОДСТВУ И ПОЙМИТЕ ИЗЛОЖЕННЫЕ В НЕМ СВЕДЕНИЯ.
- НЕСОБЛЮЖДЕНИЕ ТРЕБОВАНИЙ ДАННОГО ПРЕДУПРЕЖДЕНИЯ МОЖЕТ ПРИВЕСТИ К ТОМУ, ЧТО СПЕЦИАЛИСТ ПО ОБСЛУЖИВАНИЮ, ОПЕРАТОР ИЛИ ПАЦИЕНТ ПОЛУЧАТ УДАР ЭЛЕКТРИЧЕСКИМ ТОКОМ, МЕХАНИЧЕСКУЮ ТРАВМУ ИЛИ ДРУГОЕ ПОВРЕЖДЕНИЕ.

UPOZORENJE

(SR)

- OVO SERVISNO UPUTSTVO JE DOSTUPNO SAMO NA ENGLJESKOM JEZIKU.
- AKO KLIJENTOV SERVISER ZAHTEVA NEKI DRUGI JEZIK, KLIJENT JE DUŽAN DA OBEZBEDI PREVODILAČKE USLUGE.
- NE POKUŠAVAJTE DA OPRAVITE UREĐAJ AKO NISTE PROČITALI I RAZUMELI OVO SERVISNO UPUTSTVO.
- ZANEMARIVANJE OVOG UPOZORENJA MOŽE DOVESTI DO POVREĐIVANJA SERVISERA, RUKOVAOCA ILI PACIJENTA USLED STRUJNOG UDARA ILI MEHANIČKIH I DRUGIH OPASNOSTI.

UPOZORNENIE

(SK)

- TENTO NÁVOD NA OBSLUHU JE K DISPOZÍCII LEN V ANGLIČTINE.
- AK ZÁKAZNÍKOV POSKYTOVATEĽ SLUŽIEB VYŽADUJE INÝ JAZYK AKO ANGLIČTINU, POSKYTNUTIE PREKLADATEĽSKÝCH SLUŽIEB JE ZODPOVEDNOSŤOU ZÁKAZNÍKA.
- NEPOKÚŠAJTE SA O OBSLUHU ZARIADENIA SKÔR, AKO SI NEPREČÍTATE NÁVOD NA OBLUHU A NEPOROZUMIETE MU.
- ZANEDBANIE TOHTO UPOZORNENIA MÔŽE VYÚSTIŤ DO ZRANENIA POSKYTOVATEĽA SLUŽIEB, OBSLUHUJÚCEJ OSOBY ALEBO PACIENTA ELEKTRICKÝM PRÚDOM, DO MECHANICKÉHO ALEBO INÉHO NEBEZPEČENSTVA.

ATENCION

(ES)

- ESTE MANUAL DE SERVICIO SOLO EXISTE EN INGLES.
- SI ALGUN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLES, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCION
- NO SE DEBERA DAR SERVICIO TECNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

VARNING

(SV)

- DEN HÄR SERVICEHANDBOKEN FINNS BARA TILLGÄNGLIG PÅ ENGELSKA.
- OM EN KUNDS SERVICETEKNIKER HAR BEHOV AV ETT ANNAT SPRÅK ÄN ENGELSKA ANSVARAR KUNDEN FÖR ATT TILLHANDAHÅLLA ÖVERSÄTTNINGSTJÄNSTER.
- FÖRSÖK INTE UTFÖRA SERVICE PÅ UTRUSTNINGEN OM DU INTE HAR LÄST OCH FÖRSTÅR DEN HÄR SERVICEHANDBOKEN.
- OM DU INTE TAR HÄNSYN TILL DEN HÄR VARNINGEN KAN DET RESULTERA I SKADOR PÅ SERVICETEKNIKERN, OPERATÖREN ELLER PATIENTEN TILL FÖLJD AV ELEKTRISKA STÖTAR, MEKANISKA FAROR ELLER ANDRA FAROR.

DİKKAT

(TR)

- BU SERVİS KILAVUZUNUN SADECE İNGİLİZCESİ MEVCUTTUR.
- EĞER MÜŞTERİ TEKNİSYENİ BU KILAVUZU İNGİLİZCE DIŞINDA BİR BAŞKA LİSANDAN TALEP EDERSE, BUNU TERCÜME ETTİRMEK MÜŞTERİYE DÜŞER.
- SERVİS KILAVUZUNU OKUYUP ANLAMADAN EKİPMANLARA MÜDAHALE ETMEYİNİZ.
- BU UYARIYA UYULMAMASI, ELEKTRİK, MEKANİK VEYA DİĞER TEHLİKELERDEN DOLAYI TEKNİSYEN, OPERATÖR VEYA HASTANIN YARALANMASINA YOL AÇABİLİR.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a General Electric representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctg/InstallFulfill/InstallFulfillment.htm>
- Contact your local service coordinator for more information on this process.

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

IMPORTANT...RADIOACTIVE MATERIAL HANDLING

Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment. GE Healthcare Services is required to notify the applicable U.S. state agency PRIOR to any source service event involving pin source handling. See NUC/PET Radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GE Healthcare iTrak Process to report all omissions, errors, and defects in this publication.

REVISION HISTORY

Rev	Date	Author	Reason For Change
1	July 18 2008	Doug Hofstetter	Initial version.

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1- COMMON INSTRUCTIONS AND REQUIREMENTS

1-1 Scope

This installation specification is a control document for the installation of an Uninterruptible Power System (UPS) unit (5269966, B7864PZ, P5064PS, E4502F) to be used with General Electric Healthcare Computerized Tomography diagnostic imaging systems. This document supports CT and PET systems, which use the following Power Distribution Units:

NGPDU- 2326492x.

1-2 Overview

This manual covers the instructions for interconnection with the system (A1) power distribution panel. The term "A1" is a GE Healthcare designator for the power distribution panel that feeds power to the system. This distribution panel can be a "Main Disconnect panel" or a "Power Distribution Panel" (PDB). The Main Disconnect panel has a Knife Switch that restores/removes incoming power. The PDB uses a Rotary Switch. Throughout this manual this switch will be referred to as the Main Power Switch.

This document was designed to guide installation of a 14.4kVA UPS into CT and PET systems with the NGPDU as follows:

- PET Discovery: ST / VCT / 600 /690
- CT (7.X) VCT/ Pro32
- CT (8.X) HD 750

The UPS powers critical system computer and control functions. It is intended to bridge short power outages and provides time for crossover from normal mains power to emergency power. It is sized to provide at least 10 minutes backup with fully charged batteries.

It is recommended that if a power outage exceeds 5 minutes with no mains or emergency power, the patient should be removed from the table and an orderly shutdown of the system should be initiated.

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Panel. Use of one of the following GE panels, or equal, specifically designed for use with this kit, is recommended:

A1 Main Disconnect panels:

- E4502AB / P5051RJ (90A)
- E4502AC / P5051RK (110A)
- E4502AE (125A)
- E4502AF (150A)

PDB Panels

- CT_300
- CT_320
- CT_330

Refer to the system Pre-Installation manual for proper Catalog selection.

1-2-1 Tools Required

- FE Tool Kit
- LOTO
- Fluke 87 Multi-meter or equal
- Wire Stripper
- Labels
- Electronic Voltmeter

1-2-2 Estimated Manpower & Time Requirements

- 1 FE & 1 Electrician
- 3 hours

1-2-3 Safety Requirements

General Warning Considerations:

The only user operations permitted with the UPS are:

- Starting up and shutting down of the UPS.
- Operating the user interface in the UPS.
- Connecting data interface cables.
- Monitoring the UPS.

NOTICE

It is strongly recommended to have the User's Guide for the UPS available. These operations must be performed according to the instructions in this document and the User's Guide for the UPS being installed. This document covers installation of the POWERWARE 9355-15-14GE UPS only.

DANGER

THE UPS CONTAINS LETHAL VOLTAGES. THERE ARE NO USER SERVICEABLE PARTS INSIDE THE UPS. SERVICE AND REPAIRS MUST BE PERFORMED BY AUTHORIZED PERSONNEL ONLY.

1-3 Preliminary Instructions

Electrician

Local codes may require an Electrician to install a ¾" conduit or equivalent between the A1 Disconnect and the computer floor or to the general area of the UPS. The conduit or equivalent should be terminated in a small wall box. This will be used for the UPS Control cable connecting the UPS and A1 Main Disconnect panel or PDB panel.

Electrical Requirements

The UPS receives power from the PDU and returns power to the PDU. No other power connections are required. However, low voltage and 120Vac RPO Control and interlock signals are passed between the UPS and A1 Main Disconnect panel.

2-POWERWARE 9355-15-14GE UPS INSTALLATION

2-1 Overview

Perform the following installations and connections at the UPS.

- UPS Mechanical Preparation
- Site Preparation Verification

Verify the site has been prepared for the UPS installation and the system is in a powered down state. Mains Power should be shut off to the A1 Main Disconnect or PDB panel. Locate the Powerware User's Guide supplied with the UPS kit as necessary for additional information.

2-1-1 Inspection

The UPS is shipped bolted to a wooden pallet as shown below and protected with an outer protective wrap. Visually verify there is no evidence of shipping damage to the packaging or UPS. Remove the outer protective wrap and perform the steps in the following section to unload the UPS from the pallet.



Illustration 1: UPS on Shipping Pallet

2-2 Unloading the UPS from the Pallet

CAUTION

- The UPS cabinet is heavy. Shipping weight is approximately 300kg (650lb). If unpacking instructions are not closely followed, the cabinet may tip and cause serious injury.
- Do not install a damaged cabinet. Report any damage to the carrier and contact your Eaton service representative immediately.
- Do not tilt the UPS cabinet more than 10° from vertical or the cabinet may tip over.
- NOT use the maintenance bypass switch housing as a handle while removing the cabinet from the pallet.

1. Remove the two M10 bolts securing the stabilizing bracket to the pallet
2. Remove the four M4 screws securing the stabilizing bracket to the cabinet rear panel and remove the bracket. Retain the hardware for later use.

NOTICE

Be sure to retain the stabilizing bracket and hardware for later reassembly onto the cabinet.

3. Remove the front cover from the bottom cabinet to access the front shipping bracket. Press and release the handle latch at the bottom of the cover and then lift the cover up and off the cabinet.
4. Remove the three M10 bolts securing the rear shipping pad to the pallet and remove the shipping pad.
5. Remove the two M10 bolts securing the front shipping bracket and remove the bracket. If needed, adjust the leveling feet to release the bracket.

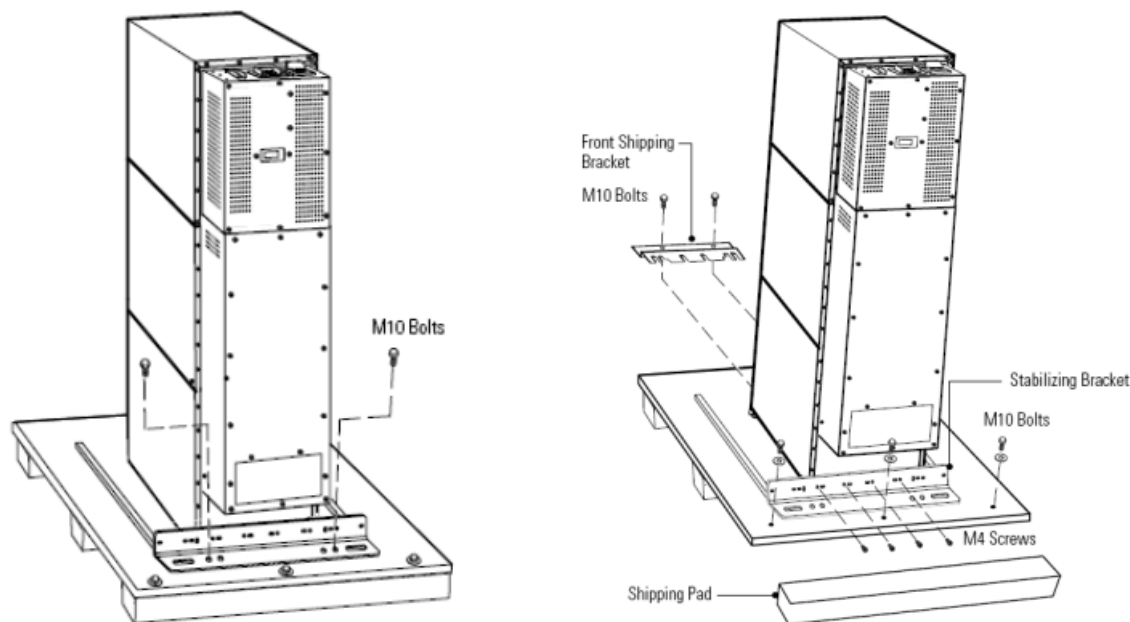


Illustration 2: Shipping Bracket Removal

6. Reinstall the front cover removed in Step 3. Hang the top edge of the cover on the cabinet first, then lower the bottom edge and snap into place.

NOTICE

In the following steps the pallet tilts and acts as a ramp once the UPS is rolled beyond the center of the pallet. Be sure to restrain the UPS as it continues to roll down the pallet / ramp. Raise feet to roll UPS, there are four in total, two in the front and two in the back.

WARNING

Do not stand directly behind the pallet while unloading the cabinet. If unloading instructions are not closely followed, the UPS may roll unexpectedly and cause serious injury.

7. With the help of two other people, making it 3 in total, slowly roll the cabinet toward the rear of the pallet. Once the pallet tilts, continue rolling the cabinet down the pallet until the cabinet touches the floor. If needed, adjust the leveling feet so that the cabinet will roll.
8. With the cabinet supported by two people, slowly pull the pallet away from the cabinet.

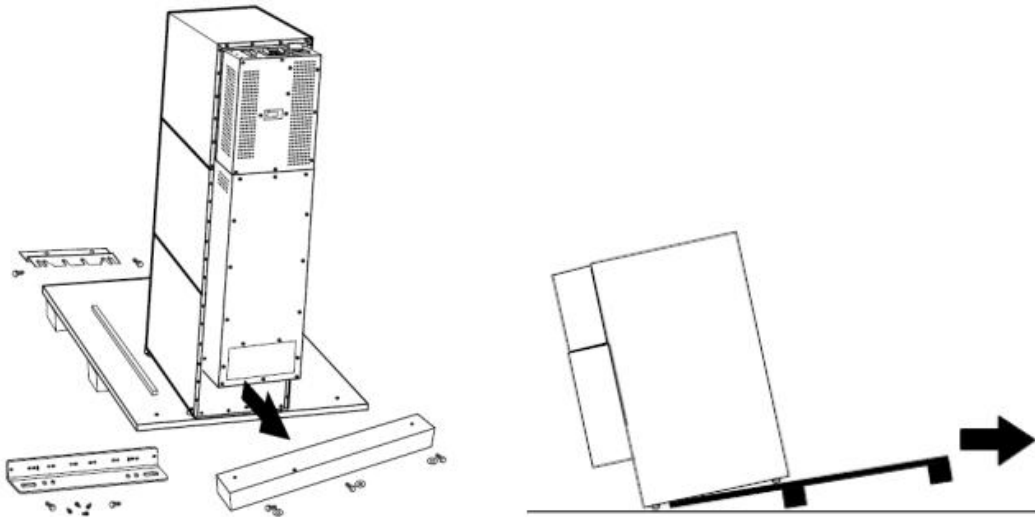


Illustration 3: Removing the Pallet

9. If installing the UPS permanently, retain the hardware for later use; otherwise, discard or recycle the pallet with the other installation shipping materials.
10. Roll the UPS to its final installation location.
11. Secure it in position by lowering the leveling feet until the UPS is not resting on the casters and it is level.
12. If a more secure installation is desired, the retained hardware may be used to secure the UPS cabinet to the floor.

2-3 Mobile Applications kit (B7864MK) – Mount Assembly

When used in mobile applications the UPS is secured with the mobile kit. The components used are supplied with the Mobile UPS kit, catalog # B7864MK.

3- POWERWARE 9355-15-14GE UPS CONNECTIONS

3-1 UPS I/O Terminal Compartment

All power and control connections are made in the I/O terminal compartment located on the rear of the UPS.

1. Verify both the UPS battery and output breakers on the top rear panel are in the “OFF” position.
2. Set the manual bypass switch to the “SERVICE” position.
3. To access the compartment, remove the two machine screws holding the cover in place and set aside for later use.
4. Remove the rear terminal box cover from the UPS.

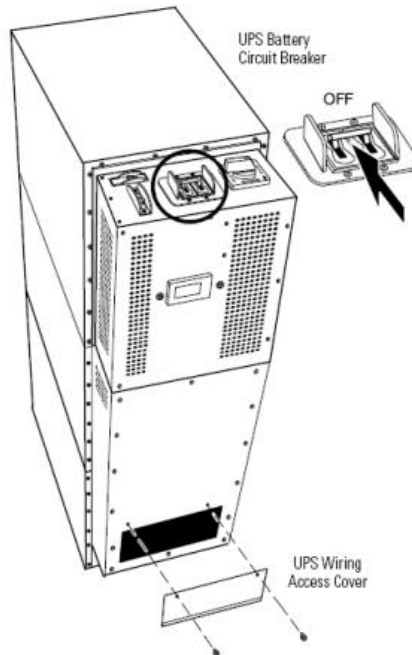
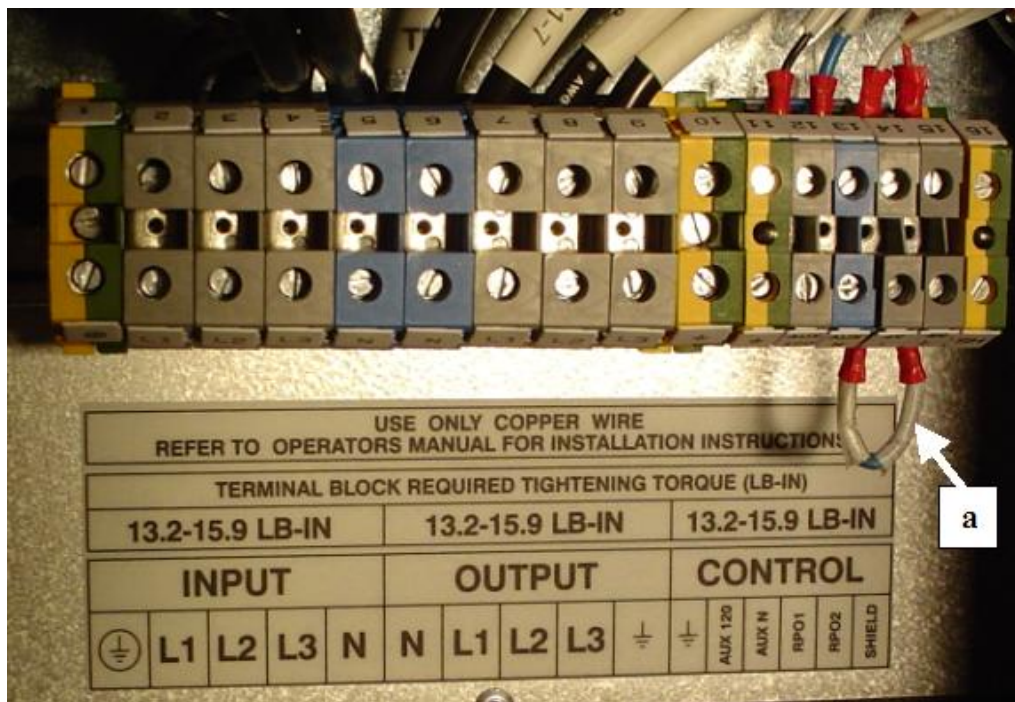


Illustration 8: UPS I/O Terminal Compartment



a- This jumper must be removed if present

Illustration 9: UPS TB1 Connections Not Cabled

5. Confirm by measurement that all terminals on the terminal strip are at 0V with respect to chassis.
6. Remove Jumper between RPO1 (Remote Power Off-1) and RPO2 (Remote Power Off-2) if present.

7. Install two 3/4" & one 1/2" cord grip fittings in rear of UPS as shown in Illustration 10. Use slip joint pliers to tighten the lock ring.
8. Use a digital voltmeter to measure the DC volts for all the connectors, connecting the ground to the green/yellow terminal. This should confirm that there is no voltage present.

3-2 UPS Input Power Cable from PDU

Refer to Illustration 10 below. Connect UPS Input Power Cable (P/N 5125079) as follows:

Cable 5125079 at UPS Power In			
Color	Terminal	Terminal Label	Cable Label
Green/Yellow	1	Ground	GND
Black	2	L1	UPS_IN A
Red	3	L2	UPS_IN B
Orange	4	L3	UPS_IN C
Blue	5	N	Neutral/N

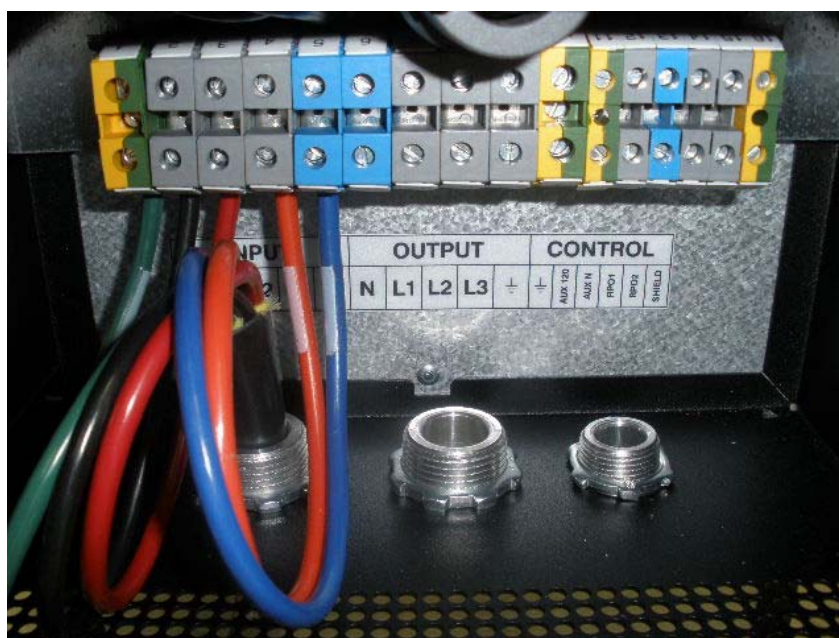


Illustration 10: UPS TB1 Connections Cabled

3-3 UPS Output Power Cable to PDU

Refer to Illustration 11 below. Connect UPS Output Power Cable (P/N 5169294-2) as follows:

Cable 5125079-2 at UPS Power Out			
Color	Terminal	Terminal Label	Cable Label
Green/Yellow	10	GND	GND
Black	7	L1	UPS_OUT A
Red	8	L2	UPS_OUT B
Orange	9	L3	UPS_OUT C
Blue	6	N	Neutral/N

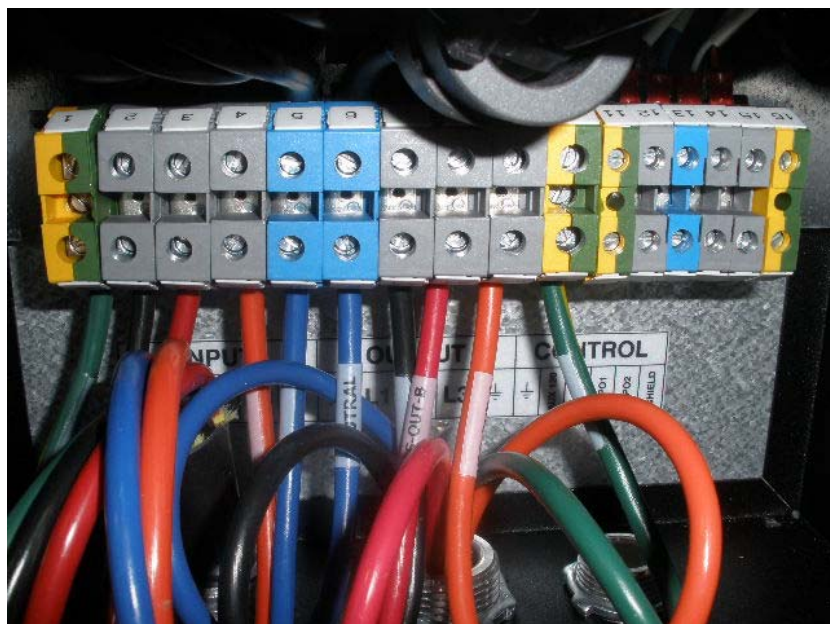


Illustration 11: UPS TB1 Connections Cabled

3-4 UPS Control Cable to A1 Main Disconnect

Refer to Illustration 12 below. Connect the UPS Control cable (P/N 5169224) as follows:

At UPS – Control cable 5169224			
Color	Terminal	Terminal Label	Cable Label
Green/Yellow	11	GND	GND
Brown	12	AUX120	AUX120
Blue	13	AUX N	NUET
Black 1	14	RPO1	RPO1
Black 2	15	RPO2	RPO2
GND	16	SHIELD	GND (SH)



Illustration 12: UPS TB1 Connections Cabled

4- POWER DISTRIBUTION UNIT

4-1 NGPDU Connections

This UPS kit has been specifically designed for compatibility with all CT and PET systems utilizing an NGPDU. Model numbers currently in use are:

- 2326492
- 2326492-2
- 2326492-3
- 2326492-4
- 2326492-5
- 2326492-6
- 2326492-7

The same interconnection information applies to all listed NGPDU models.

NOTICE

Use all required applicable safety Lockout/Tagout procedures and processes.

4-2 UPS Input Power Cable From PDU

1. If not done previously, shutdown system software and turn off the mains input power to the system.
2. Remove the top and front covers of the PDU.
3. Remove all three jumper wires at terminal strip TS4, terminals 1 to 6, 2 to 7, & 3 to 8. Place the jumpers in the base of the PDU for future use.
4. Route the UPS Input Power Cable (P/N 5125079) into the PDU and up through the indicated cable clamp in the bottom pan. Refer to Illustration 13 below.
5. Connect the wires as follows:

At PDU to UPS Power In		
Color	Terminal	Label
Green/Yellow	5	TS4-5
Black	1	TS4-1
Red	2	TS4-2
Orange	3	TS4-3
Blue	4	TS4-4

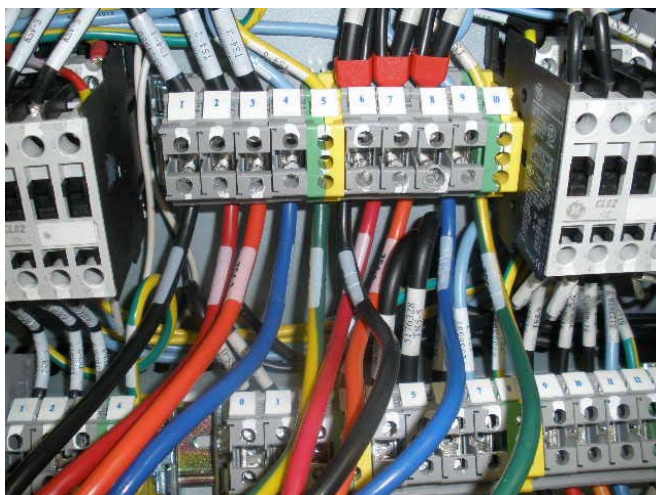


Illustration 13: NGPDU Connections

4-3 UPS Output Power Cable

1. Route the UPS Output Power Cable (P/N 5169294-2) into the PDU and up through the indicated cable clamp in the bottom pan.
2. Connect the wires as follows:

At PDU Power Out from UPS		
Color	Terminal	Label
Green/Yellow	10	TS4-10
Black	6	TS4-6
Red	7	TS4-7
Orange	8	TS4-8
Blue	9	TS4-9

NOTICE

Do not re-install the PDU covers at this time. Wait until after the system start-up procedures have been completed.

5- DISCONNECT PANEL

5-1 Option 1- Main Disconnect Panel

Use of one of the following GE panels, or equal, specifically designed for use with this kit, is recommended. Refer to your system Pre-Installation manual for proper selection.

- E4502AB / P5051RJ (90A)
- E4502AC / P5051RK (110A)
- E4502AE (125A)
- E4502AF (150A)



Illustration 14: Typical A1 Panel Front View & Interior View.



THE FOLLOWING PROCEDURES ARE ACCOMPLISHED INSIDE THE A1 DISCONNECT PANEL. MAINS VOLTAGE MAY STILL BE PRESENT EVEN WITH THE MANUAL SWITCH PULLED. LOCATE THE FEEDER DISCONNECT TO THE A1 PANEL AND SHUT OFF MAIN POWER THERE. USE PROPER LOCK-OUT / TAG-OUT PROCEDURES. TEST THE A1 DISCONNECT TO VERIFY THAT ALL POWER IS OFF BEFORE PROCEEDING.

5-1-1 UPS Control Cable Connections to Main Disconnect Panel

If required by local codes, an Electrician may be required to install 3/4" conduit or raceway between the Main Disconnect Panel and the floor in the general area of the UPS. Conduit should be terminated in a small wall box. This route will be used for the UPS control cable connecting the UPS and (A1) Main Disconnect panel.

NOTICE

Verify the feeder to the A1 Panel has been shut off and properly locked out.

CAUTION

Remove the factory-installed jumper between terminals TB1-7 and TB1-9 in the A1 Panel (Main Disconnect) if present. This is necessary to prevent damage to the UPS EPO Circuitry.

1. Install a 1/2" cord grip (supplied with the kit) in the wall box or the base of the A1 Main Disconnect Panel.
2. Route the UPS Control Cable (P/N 5169224) from the UPS through the conduit or raceway to the A1 Main Disconnect Panel.
3. Connect the BLK 1 wire to terminal TB1-7 in the A1 Panel. Look at the insulation of the wire to determine which black wire to use.
4. Connect the BLK 2 wire to terminal TB1-8 in the A1 Panel.
5. Connect the BRN wire to terminal TB1-9 in the A1 Panel.
6. Connect the BLU wire to terminal TB1-10 in the A1 Panel.
7. Connect the GREEN/YELLOW (GND) wire and the SHIELD green/yellow wire to the Ground Terminal (TB1-11) in the A1 Panel.

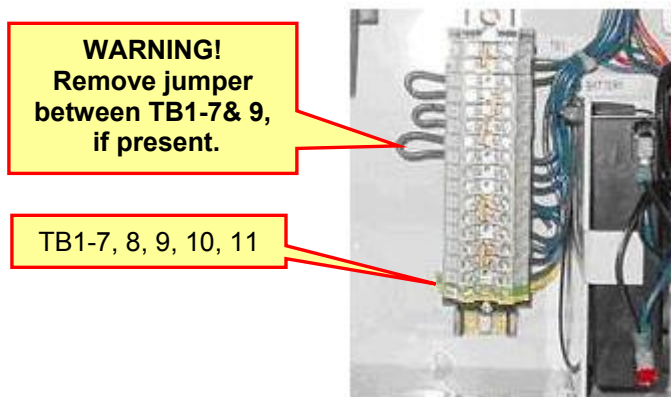


Illustration 15: A1 Panel TB1 Terminals

5-1-2 A1 Panel Voltage Selection

1. Verify the A1 Main Disconnect Panel control transformer is configured for the correct nominal mains voltage. The panel ships from the factory configured for 480V input. Reconnect for 400V if required.

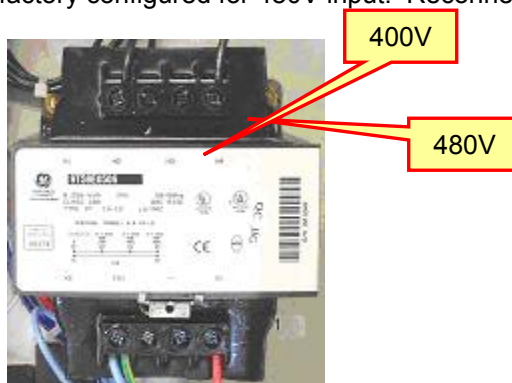


Illustration 16: A1 Panel Control Transformer

5-1-3 Main Disconnect Panel Labeling

A pre-printed adhesive “SYSTEM ON AND SYSTEM OFF” label is provided in the UPS kit. Attach one of the labels provided in the Hardware Kit to the middle of the A1 Main Disconnect panel as indicated below.



Illustration 17: A1 Panel Labeling

5-2 Option 2- PDB Main Disconnect (Typical European Installations)

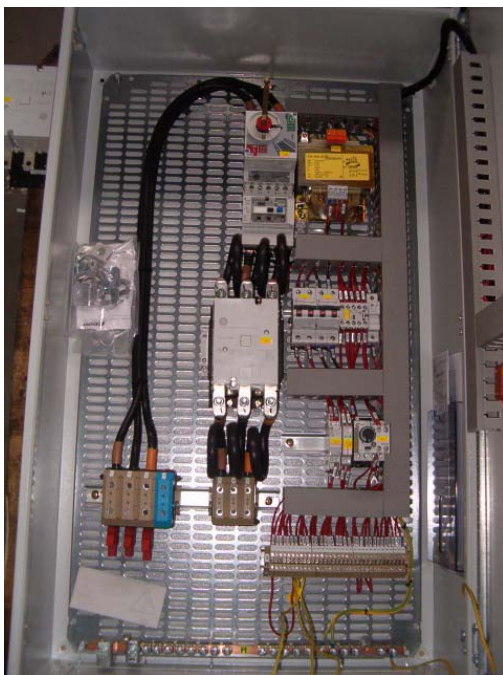
5-2-1 UPS Control Cable Connections to PDB Panel

If required by local codes, an electrician may be required to install 3/4” conduit or raceway between the PDB panel and the floor in the general area of the UPS. Conduit or equivalent should be terminated in a small wall box. This route will be used for the UPS control cable connecting the UPS and PDB panel.

NOTICE

Verify the feeder to the PDB Panel has been shut off and properly locked out.

1. Install a ½ inch cord grip (supplied with the kit) in the wall box or the base of the A1 PDB panel.
2. Route the UPS Control Cable (P/N 5169224) from the UPS through the conduit or equivalent, or raceway to the A1 PDB panel.
3. Connect the BLK 1 wire to terminal TB-26 in the PDB Panel.
4. Connect the BLK 2 wire to terminal TB-27 in the PDB Panel.
5. Field Insulate the BLK-3 or BRN (120V) wire. **For safety, it must be terminated.**
6. Field Insulate the BLK-4 or BLUE (Neutral) wire. **For safety, it must be terminated.**
7. Connect the GREEN/YELLOW (GND) and SHIELD wires to PDB Chassis Ground



Connect Ground
and Shield Cables

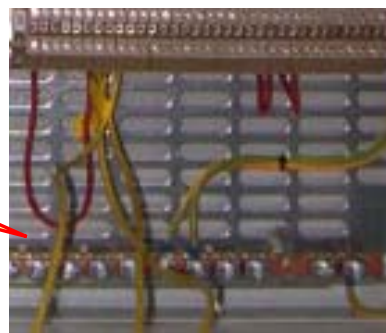


Illustration 18: Inside of PDB Panel and Ground Terminals

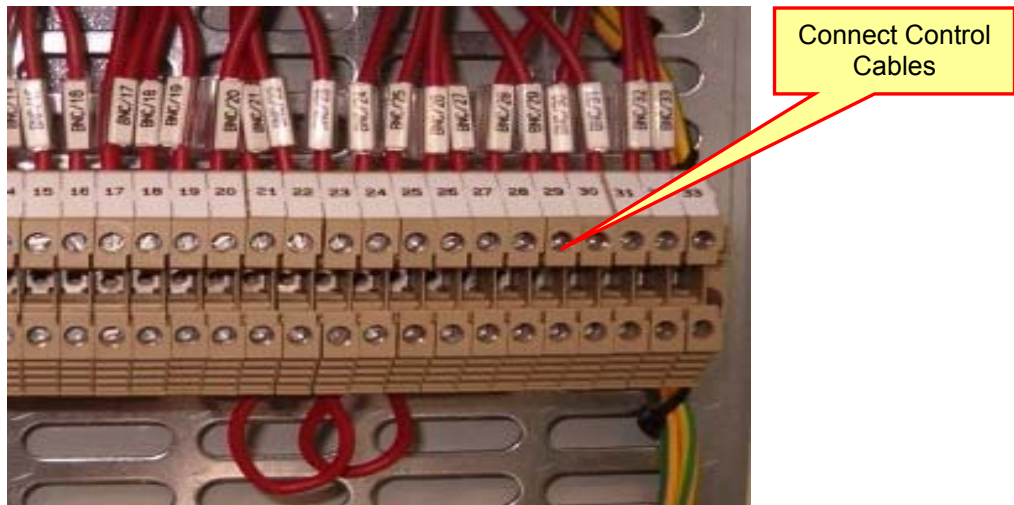


Illustration 19: PDB I/O Cable Terminals

5-2-2 PDB Panel Labeling

A pre-printed adhesive “SYSTEM ON AND SYSTEM OFF” label is provided in the UPS kit. Attach one of the labels provided in the Hardware Kit to the middle of the PDB Panel as indicated below.

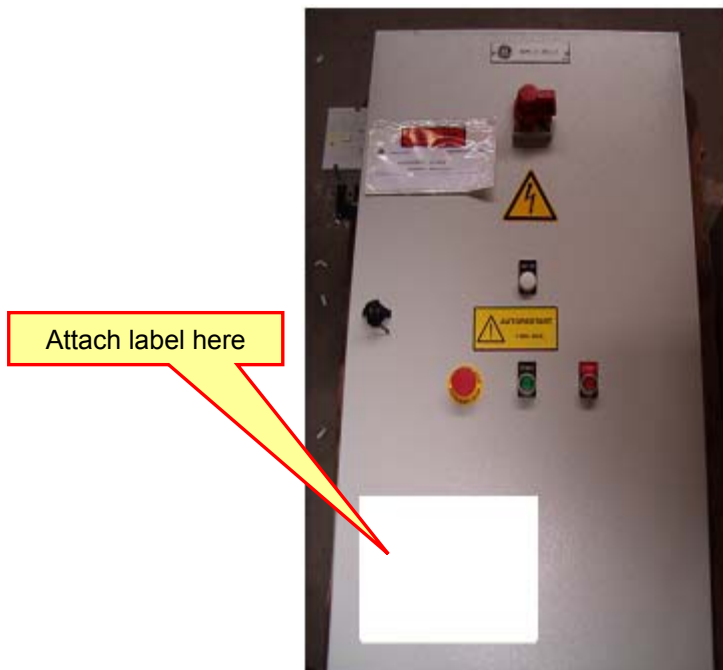


Illustration 20: PDB Panel Labeling

5-3 UPS Labeling

A pre-printed adhesive “SYSTEM ON AND SYSTEM OFF” label is provided in the UPS kit. Attach the label to the top of the UPS Panel as indicated below.

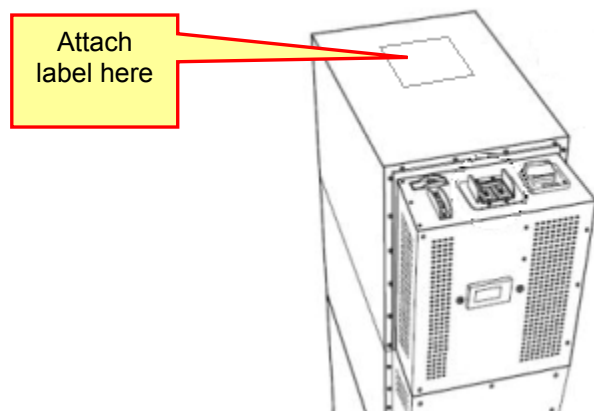


Illustration 21: UPS Labeling

6- START UP / SHUTDOWN PROCEDURES

NOTICE

GEHC installation team and service employees are authorized to perform initial power-up of the Powerware 9355-15-14GE UPS. It is not necessary to call a Powerware representative for initial start-up. However, if you prefer or have difficulty with the UPS start-up procedure, you may arrange for a Powerware service technician to perform start-up, at no cost. The Powerware contact number in the US is 1-800-843-9433. For international service refer to the contact information provided with the Powerware documentation.

6-1 Start-Up: Powerware 9355-15-14GE

6-1-1 Automatic Configuration Mode

Automatic configuration is performed during UPS startup. The UPS will automatically sense the input voltage and frequency and match its output to the mains input power. **No additional frequency or voltage configuration set-up is required.**

6-1-2 UPS Start-up Sequence



Illustration 22

The following procedure should be used to power up the system up for the first time after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off.

1. Turn OFF the Console power switch. This keeps the system computers from needlessly booting up until proper UPS system operation is verified.
2. The Mains input power was turned off prior to starting this procedure. Restore incoming power as required at the feeder panel supplying the A1 Main Disconnect or PDB Panel.

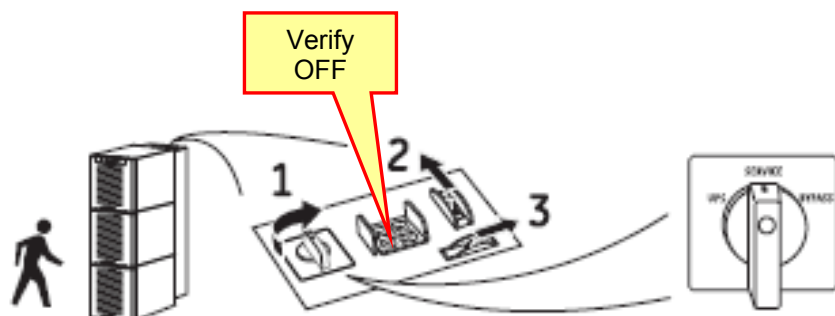


Illustration 23

3. Set the Manual By-Pass Switch to the “Service” position.
4. Verify the Output and Auxiliary circuit breakers on the UPS are in the “ON” position and the Battery circuit breaker is “OFF”.

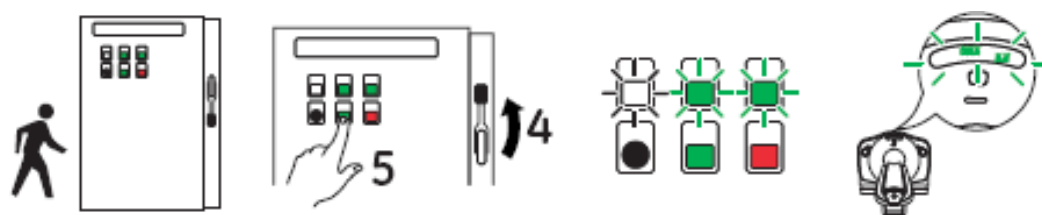
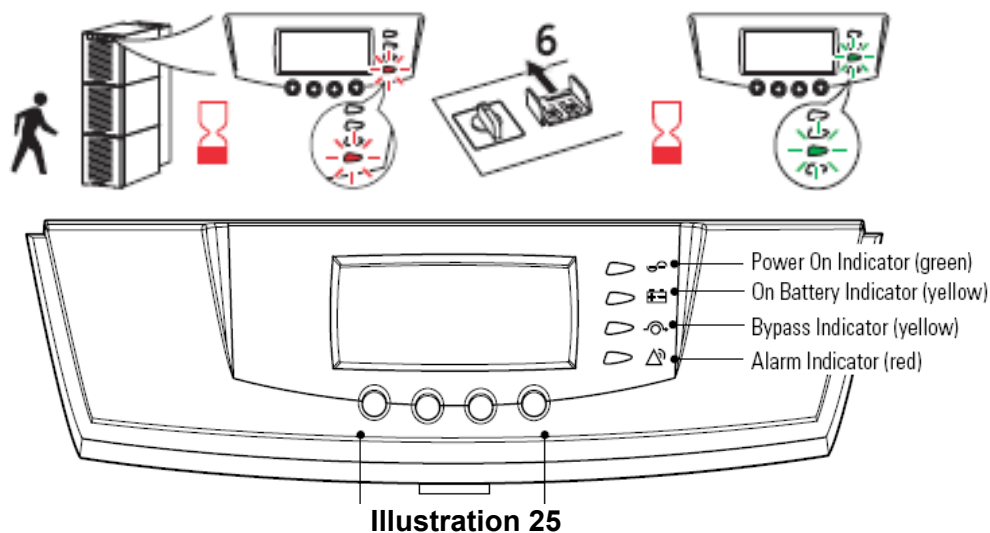
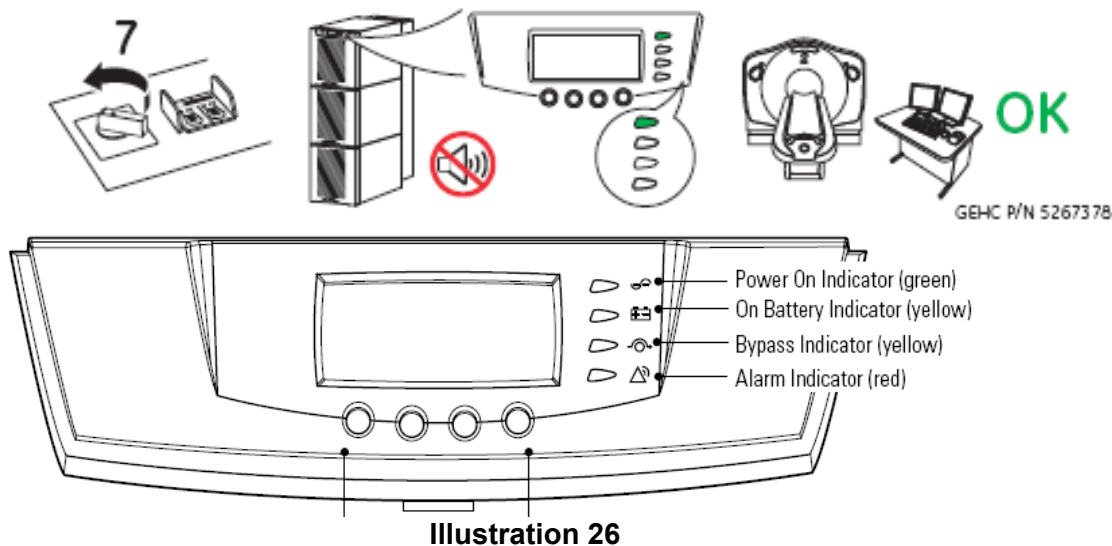


Illustration 24

5. Some Main Disconnects have a “Control Power” On/Off, key switch. If necessary, turn the Key to the ON position.
6. Place the Main Power Switch at the A1 Main Disconnect Panel or PDB, to the “ON” position.
7. Momentarily press the “START” button on the A1 panel.
8. After approximately 10 seconds, verify ‘CONTACTOR ON’ and “UPS ON” lights on the A1 Main Disconnect Panel are illuminated.
9. Verify both the UPS and scanner begin to power up.
10. At the Service Switch panel in the Gantry press “ESTOP RESET” button or press any Emergency Stop Reset button on one of the Gantry Operator panels.
 - a. If the Gantry or system fails to power on:
 1. Ensure that the Axial Enable and 120 VAC switches at the Gantry Service Switch Panel within the Gantry is set to “ON” or enabled position.
 2. Ensure the PDU is powered ON.
 3. Ensure that none of the rooms Emergency Off (SEO) buttons are latched open.



11. After approximately 10 seconds, the UPS "Alert Indicator LED" should turn on RED.
12. Re-set the UPS Battery breaker to the "ON" position.
13. After approximately 30 seconds, the UPS "By-pass Indicator" should display steady YELLOW.



14. Switch the Manual By-pass switch to the "UPS" position.
15. After a few seconds, the UPS "Power On Indicator" should turn GREEN (Steady) and the UPS should stop making audible "beeps".
16. Proceed to Section 6-2 "Complete System and UPS Shutdown Test".

NOTICE

The UPS is now providing 120VAC power to the Table, Gantry and Console. In the event of a momentary mains power loss, scanning will be inhibited or aborted. However, the UPS will continue to provide power to the Table, Gantry & Console. The UPS will “beep” occasionally, letting you know that it is on BATTERY BACKUP. If mains power is restored prior to shutting down the CT system, the UPS and system will automatically return to normal operation.

6-1-3 UPS Alarm Conditions

Input Phase Rotation Condition: If the UPS does not start properly and the display indicates a “Phase Rotation” message, there is an input phase error. Perform LOTO and Swap L1 and L2 at the PDU Mains Input. Do **NOT** reverse any of the connections at the UPS or PDU TS4 Interface.

Other Alarms

Refer to the 9355-15-14GE User’s Guide for a detailed listing of operational and diagnostic information for the UPS.

NOTICE

The system cannot be turned over to the Customer unless the UPS has been tested and has been determined to be fully functional.

6-2 Complete System and UPS Shutdown Test

This function removes power to the entire system including the UPS.

6-2-1 System and UPS Shutdown Using Emergency OFF

NOTICE

This section must be performed with the console OFF. Make sure that the system software is shutdown normally to avoid software corruption and the console power switch is in the “OFF” position.



Illustration 27

Test the function of the UPS mode of operation as follows:

1. Verify the system software is shutdown and the Console power switch is “OFF”.
2. At the A1 main Disconnect Panel, press the “Emergency OFF” button.
 - Verify the A1 Main Disconnect or PDB panel shuts off immediately.
 - Verify the UPS drops the load immediately.
 - Verify the system and UPS completely shuts down within approximately one minute and the system and UPS power is off.

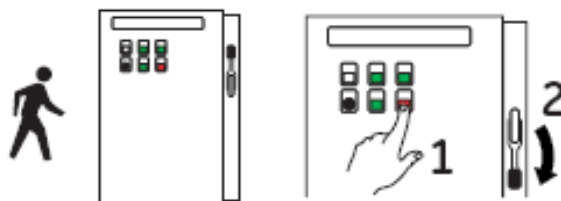


Illustration 28

3. Some Main Disconnects have a “Control Power” On/Off, key switch. If necessary, turn the Key to the “OFF” position.
4. Place the Main Power Switch at the A1 Main Disconnect Panel or PDB, to the “OFF” position.

6-2-2 Restore Power

1. Set the Manual By-Pass Switch to the “Service” position.
2. Verify the Output and Auxiliary circuit breakers on the UPS are in the “ON” position and the Battery circuit breaker is “OFF”.

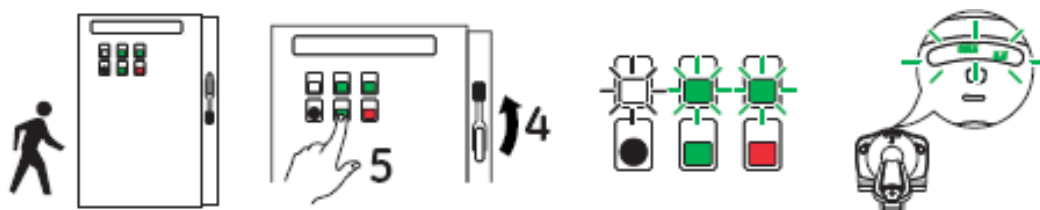
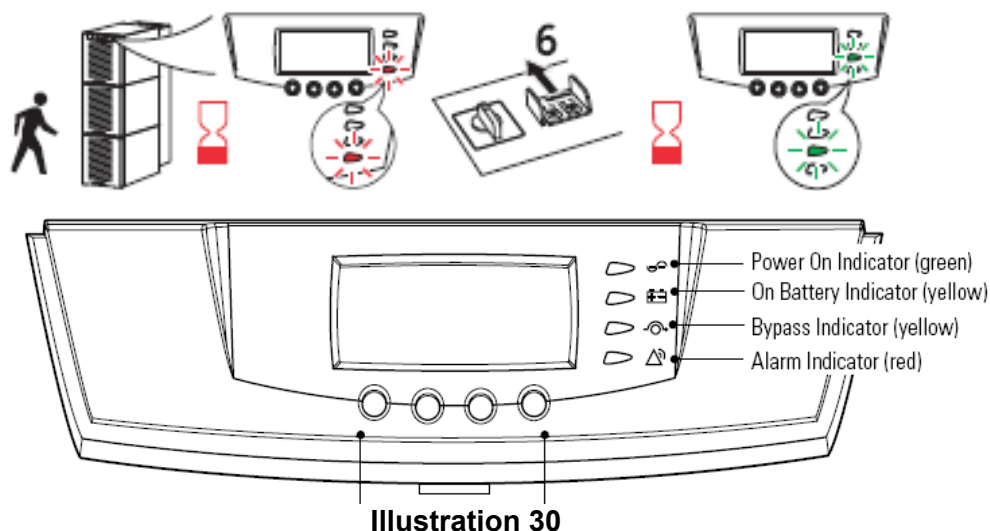
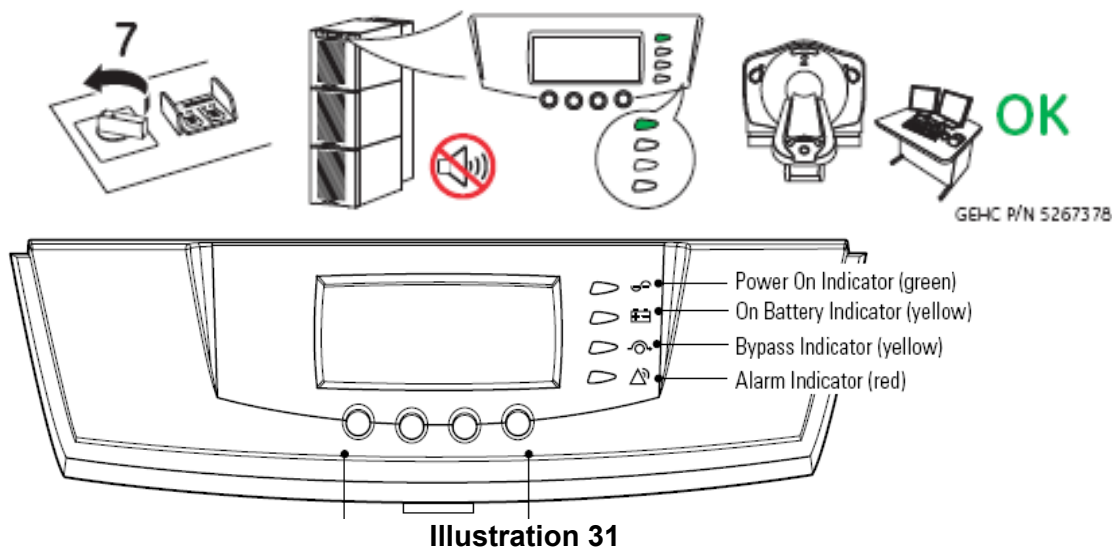


Illustration 29

3. Some Main Disconnects have a “Control Power” On/Off, key switch. If necessary, turn the Key to the ON position.
4. Place the Main Power Switch at the A1 Main Disconnect Panel or PDB, to the “ON” position.
5. Momentarily press the “START” button on the A1 panel.
6. After approximately 10 seconds, verify ‘CONTACTOR ON’ and “UPS ON” lights on the A1 Main Disconnect Panel are illuminated.
7. Verify both the UPS and scanner begin to power up.
8. At the Service Switch panel in the Gantry press “ESTOP RESET” button or press any Emergency Stop Reset button on one of the Gantry Operator panels.
 - a. If the Gantry or system fails to power on:
 4. Ensure that the Axial Enable and 120 VAC switches at the Gantry Service Switch Panel within the Gantry is set to “ON” or enabled position.
 5. Ensure the PDU is powered ON.
 6. Ensure that none of the rooms Emergency Off (SEO) buttons are latched open.



9. After approximately 10 seconds, the UPS "Alert Indicator LED" should turn on RED.
10. Re-set the UPS Battery breaker to the "ON" position.
11. After approximately 30 seconds, the UPS "By-pass Indicator" should display steady YELLOW.



12. Switch the Manual By-pass switch to the "UPS" position.
13. After a few seconds, the UPS "Power On Indicator" should turn GREEN (Steady) and the UPS should stop making audible "beeps".
14. To proceed:
 - a. If the UPS is to remain engaged, (Providing partial UPS protection) proceed to section 6-4 "Customer Training-Translation of Symbols".
 - b. If the UPS is to be disengaged (Turned off) proceed to section 6-3 "UPS By-pass Operation".

6-3 UPS By-pass Operation

This section allows for the normal shutdown and operation of the system without any UPS functionality.

6-3-1 UPS By-pass Mode

To run the system in normal operation without the UPS capability, do the following:

1. Turn the Manual Bypass switch to the "BYPASS" position.
 - The UPS "Battery" circuit breaker will trip almost immediately.
 - After approximately 1 minute the UPS should automatically shut down.
2. Verify the UPS "Battery" circuit breaker is in the "OFF" position.
3. This completes the shutdown of the UPS.
4. Proceed to the next section. (6-4 Customer Training-Translation of Symbols).



ALTHOUGH THE UPS STOPS SUPPLYING POWER AND IS DISCONNECTED INTERNALLY FROM ITS BATTERIES, SYSTEM POWER STILL PASSES THROUGH THE UPS CABLES WHILE IN BYPASS MODE. USE CAUTION WHEN SERVICING.

6-4 Customer Training - Translation of Symbols

The purpose of this section is to familiarize the responsible site personnel with the shutdown and power up procedures for the A1 Main Disconnect/PDB panel, UPS and system so they are familiar with the symbolic labels that describe the functions of these devices.

1. Turn on the "Power" button at the console and allow the system software to boot up normally.
2. Proceed to section 6, sub-section 6-1 "Start-Up: Powerware 9355-15-14GE", to restore power to the UPS.
3. At this time, consult with the responsible site person or persons to make sure they understand the labels that define the Startup and Shutdown procedures.

6-5 Final Steps

1. Re-install all safety shields and covers on the UPS and PDU.
2. Fill out and mail the Powerware Warranty Registration card or register the UPS on-line at www.warranty.powerware.com.
3. *For Europe and EAGM Pole Only:* Call Eaton in UK + 44 (0) 845-177-3410 or send a mail to GEMedSvcReqs@eaton.com to register and warrant the unit. The UPS serial number and hospital address are required. A warranty contract number will then be generated by Eaton. This number is needed to request any Eaton intervention during the warranty period.

7- APPENDIX A

A-1 Applicable Documents

The following documents provide reference information for the UPS Interconnect Kit.

- GEHC Reference Documents 5269966SCH UPS Interconnect Schematic

A-2 Supplier Documents

POWERWARE 9355-15-14GE User's Guide

A-3 Field Replaceable Units (FRU) Listing

The following listing identifies the Field Replaceable Units (FRUs) associated with this assembly. Vendor part numbers are included as examples only. For the complete list of approved vendors for each item, see the associated Manufacturer's Equivalent on GEHC ePDM system.

GEMS Part #	Name / Description	Vendor	Vendor P / N
5198342	PW9355-15-14GE UPS	EATON, POWERWARE	103006185
5269476	Hardware Kit	GEXPRO	5269476
5308485	Long Cable Kit* (Optional)	GEXPRO	5308485

*If longer power and communication cables (19 meters each) are needed, order the Long Cable Collector Kit.

8- APPENDIX B – PARTS DISPOSITION AND RETURN

The customer has purchased this upgrade. Legally all new parts added and the old parts removed are the property of the customer. However, please ask the customer if they would like to keep the old parts, or have you dispose of the old parts.

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